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Airbus Material Specification

Carbon Fiber Reinforced Epoxy Prepreg Unidirectional Tape / 180 °C Curing Class / High Performance Resin

Intermediate Modulus Fiber Structural Material

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1 Scope

This Material Specification defines the requirements for a group of materials each consisting of a carbon fiber reinforced high performance epoxy resin UD tapes prepregs system, curing at 180 °C, for aerospace application.

The requirements specified in this Material Specification shall not be used as stress allowable.

Unless otherwise specified in the drawings, the purchase order or the inspection schedule, the presented specification shall be used in conjunction with the referenced documents.

2 Application

This material is used for manufacturing primary airframe and aerospace structures (sandwich and monolithic). The material is to be delivered according to the relevant ABS5377.

3 Normative References

This Airbus Standard incorporates by dated or undated reference provisions from other publications. All normative references cited at the appropriate places in the text are listed hereafter. For dated references, subsequent amendments to or revisions of any these publications apply to this Airbus Standard only when incorporated in it by amendment of revision. For undated references, the latest issue of the publication referred to shall be applied.

ISO1183	Plastics - Methods for determining the density and relative density of non-cellular plastics
ISO10119	Carbon fibers - Determination of density
ISO11359-2	Plastics - Thermomechanical analysis (TMA). Determination of coefficient of linear thermal expansion and glass transition temperature
EN2378	Aerospace series - Fiber reinforced plastics; Determination of water absorption by immersion
EN2557	Aerospace series - Carbon fiber preimpregnates; Determination of mass per unit area
EN2558	Aerospace series - Carbon fiber preimpregnates; Determination of the percentage of volatile matter
EN2559	Aerospace series - Carbon fiber preimpregnates; Determination of the percentage of resin and fiber content and the mass of fiber per unit area
EN2560	Aerospace series - Carbon fiber preimpregnates; Determination of the resin flow
EN2561	Aerospace series - Carbon fiber reinforced plastics unidirectional laminates; Tensile test parallel to the fiber direction
EN2563	Aerospace series - Unidirectional laminates; Determination of the apparent inter laminar shear strength
EN2597	Aerospace series - Unidirectional laminates carbon thermosetting resin; Tensile test perpendicular to the fiber direction
EN2823	Aerospace series - Fiber reinforced plastics; Determination of the effect of exposure to humid atmosphere on physical and mechanical characteristics
EN2850	Aerospace series - Carbon fiber thermosetting resin unidirectional laminates compression test parallel to the fiber direction
EN3615	Aerospace series - Fiber reinforced plastics; Determination of the conditions of exposure to humid atmosphere and moisture absorption
ABS5377	Product Standard - High Performance Carbon fibre reinforced epoxy prepreg. Unidirectional tape / 180 °C curing class / High performance resin / Intermediate modulus fibre; Structural material

AIMS05-27-000 Part 1	Technical Specification – High performance carbon fiber reinforced epoxy prepreg; Part 1: General Requirements
AIMS05-27-000 Part 2	Technical Specification - High performance carbon fiber reinforced epoxy prepreg; Unidirectional tape / 180 °C curing class; Structural material; Part 2: Qualification program / batch release testing and incoming inspection testing
AIMS05-27-000 Part 100	Technical Specification - High performance carbon fiber reinforced epoxy prepreg – Structural Material - Acceptance Testing and storage life prolongation
AITM1-0001	Test Method - Fiber reinforced plastics; Determination of mechanical degradation due to chemical paint strippers
AITM1-0002	Test Method - Fiber reinforced plastics; Determination of in-plane shear properties
AITM1-0003	Test Method - Fiber reinforced plastic materials; Determination of the glass transition temperatures
AITM1-0005	Test Method - Carbon fiber reinforced plastics; Determination of inter laminar fracture toughness energy; Mode I
AITM1-0006	Test Method - Carbon fiber reinforced plastics; Determination of inter laminar fracture toughness energy; Mode II
AITM1-0007	Test Method - Fiber reinforced plastics; Determination of plain, open hole and filled hole tensile strength
AITM1-0008	Test Method - Fiber reinforced plastics; Determination of plain, open hole and filled hole compression strength
AITM1-0009	Test Method - Fiber reinforced plastics; Determination of bearing strength
AITM1-0010	Test Method - Fiber reinforced plastics; Determination of compression strength after impact
AITM3-0001	Test Method - Analysis of thermoset systems by High performance liquid chromatography
AITM3-0002	Test Method - Analysis of non-metallic materials (uncured) by Differential scanning calorimetry
AITM3-0003	Test Method - Analysis of organic compounds by infrared spectroscopy
AITM3-0004	Test Method - Determination of gel time and viscosity
AITM3-0008	Test Method - Determination of the extent of cure by differential scanning calorimetry
AITM6-0011	Test Method - Non Destructive Inspection of Composite Parts
AIPS03-02-019	Process Specification - Manufacture of Monolithic Parts with Thermoset Prepreg Materials
AIPS03-07-019	Process Specification - Forming of Thermoset Prepreg Materials
AIPS03-08-003	Process Specification - Rework and repair of Structures Manufactured from Composite Materials (Laminates and Sandwich)
AIPS06-01-003	Process Specification - Surface Preparation for Thermosetting parts before structural bonding
AIPS06-02-006	Process Specification - Structural Bonding of Thermoset and Thermoplastic Matrices; Composite Parts
ASTME831	Standard test method for linear thermal expansion of solid materials by thermomechanical analysis

4 Requirements

4.1 General

Specification values are minimum except where stated otherwise.

4.1.1 Material Description

See Table 1.

Table 1: Material Description

Characteristic	Requirements		
Type	UD carbon fiber tape/epoxy resin		
Formulation	Fibre: Intermediate modulus carbon fibre Resin: Epoxy, toughened		
Form and method of production	Prepreg, cured in an autoclave at 180 °C		
Technical Specification	AIMS05-27-000 Part 1 and Part 2		
Material Code	A	B	C
Nominal Fiber Areal Weight (FAW)	134 g/m ²	194 g/m ²	268 g/m ²
Cured Ply Thickness (CPT) used for normalisation	0,125 mm	0,181 mm	0,250 mm

4.1.2 Work life and storage conditions

See Table 2.

Table 2: Work life and storage conditions

No.	Properties	Shop condition	Symbol	Unit	Requirements
1	Storage life at -18 °C	-	-	months	≥ 12
2	Storage life at max +5 °C	-	-	months	a)
3	Work life	27 °C / 70 % R.H.	-	days	≥ 10
		24 °C / 60 % R.H.	-	days	≥ 30
a) To be reported during qualification					

4.1.3 Processing

The Material shall meet the technical requirements of Airbus applicable documents (including AITM6-0011 "Airbus Test Method for Inspection Processes Non Destructive Inspection of Composite Parts") for composite parts standard manufacturing process for aerospace related applications effective as at the date of qualification Status A and Status T of the Material, as applicable to the Purchasers and the Purchasers' enabled Suppliers, including the following Airbus process specifications:

- AIPS03-02-019 "Manufacture of monolithic parts with thermoset prepreg materials"
- AIPS03-07-019 "Forming for thermoset prepreg materials"
- AIPS03-08-003 "Rework of structures manufactured from composite materials (laminated and sandwich)"
- AIPS06-01-003 "Surface preparation for thermosetting parts before structural bonding"
- AIPS06-02-006 "Structural bonding of thermoset and thermoplastic matrices composite parts"

The curing cycle shall be agreed between the manufacturer and Airbus at the onset of the material qualification program and shall be used to produce all test specimens for qualification testing. The laminates for test purposes shall be cured in an autoclave. The curing cycle shall comply with the general requirements reported in Table 3. The agreed curing cycle shall be specified in the relevant IPS and shall be used for the manufacturing of batch release test specimens.

Table 3: Processing conditions

Cure type	Autoclave		
Cure conditions	Heating rate	°C/min	0,2 - 5,0
	Cure temperature	°C	180 ⁺¹⁰ / ₋₅
	Cure time	Min	120 - 240
	Cure pressure	MPa	0,15 - 1,1
		Bar	1,5 - 11
	Cooling rate	°C/min	0,2 - 3,5
	Vacuum (abs)	MPa	0,005 - 0,1
NOTE: The use of an intermediate temperature and low-pressure dwell is permissible if agreed between manufacturer and purchaser. Refer to relevant IPS for detailed cure cycle to be applied.			

4.2 Specific**4.2.1 Physical-Chemical Properties**

Uncured and cured properties see Table 4.

Table 4: Physical-chemical properties

Property		Test method	Unit	Requirement		
Material code			AIMS05-27-002			
			A	B	C	
Uncured						
Prepreg mass/unit area		EN2557C	g/m ²	203 ± 7	294 ± 9	406 ± 11
Fibre mass/unit area		EN2559	g/m ²	134 ± 3	194 ± 5	268 ± 7
Resin density		ISO1183 Method A	g/cm ³	a)		
Fibre density		ISO10119	g/cm ³	1,78 ± 0,04		
Volatile content		EN2558	%	< 2		
Resin content		EN2559	%	34 ± 1		
Physic-chemical tests	HPLC	AITM3-0001	-	a)		
	DSC	AITM3-0002	-	a)		
	IR	AITM3-0003	-	a)		
	Gel Time	AITM3-0004 ethod B, C or E	-	a)		
	Viscosity	AITM3-0004 Method A	-	a)		
Resin flow		EN2560	%	> 2		
Tack		To be agreed between manufacturer and Airbus	-	a)		
Cured						
Coefficient of thermal expansion		ASTME831 ^{b)}	-	c)		
Water uptake		EN3615, 70 °C/85 % R.H.	%	< 1,4		
Water uptake		EN2378, 70 °C/water	%	c)		
Extent of cure		AITM3-0008	%	c)		
NOTE: The stated requirements are average values.						
a) Equipment settings and value to be agreed between manufacturer and Airbus.						
b) It is also allowed to use ISO11359-2.						
c) Results to be reported.						

4.2.2 Mechanical Properties

See Table 5 and Table 6.

Table 5: Mechanical Properties

No	Property	Test method	Unit	Conditioning ^{b)}	Test temp. [°C]	Requirement ^{a)}	
						X	X _{min}
Material Code						AIMS05-27-002 A, B and C	
1	Glass transition temperature, T _{g onset}	AITM1-0003	°C	dry	-	-	177
				wet	-	-	120
2a	Tensile strength 0°	EN2561 Type B	MPa	dry	-55	2790	2750
					RT	2770	2650
					70	2550	2500
				wet	70	2550	2500
					90	2550	2500
2b	Tensile modulus 0°	EN2561 Type B	GPa	dry	-55	182 ± 12	182 ± 15
					RT	182 ± 12	182 ± 15
					70	182 ± 12	182 ± 15
				wet	70	182 ± 12	182 ± 15
					90	182 ± 12	182 ± 15
2c	Poisson Ratio	EN2561 Type B	GPa	dry	RT	-	
				wet	70	-	
					90	-	
3a	Tensile strength 90°	EN2597 Type B	MPa	dry	-55	45	40
					RT	50	45
				wet	70	33	30
					90	28	25
3b	Tensile modulus 90°	EN2597 Type B	GPa	dry	-55	9,1 ± 0,4	
					RT	9,0 ± 0,4	
				wet	70	7,7 ± 0,4	
					90	7,2 ± 0,4	
4a	Compression strength 0°	EN2850 Type B	MPa	dry	-55	1400	1270
					RT	1400	1270
					90	1400	1270
				wet	70	1210	1100
					90	1130	1020
(continued)							

(continued)

Table 5: Mechanical Properties

No	Properties	Test method	Unit	Conditioning ^{b)}	Test temp. [°C]	Requirements ^{a)}	
						X	X _{min}
Material Code						AIMS05-27-002 A, B and C	
4b	Compression modulus 0°	EN2850 Type B	GPa	dry	-55	150 ± 10	
					RT	150 ± 10	
					90	150 ± 10	
				wet	70	150 ± 10	
					90	150 ± 10	
5a	Compression strength 90°	EN2850 Type B	MPa	dry	RT	270	260
				wet	70	180	170
					90	150	140
5b	Compression modulus 90°	EN2850 Type B	GPa	dry	RT	9,7 ± 0,3	
				wet	70	8,4 ± 0,3	
					90	7,9 ± 0,3	
6a	In plane shear Strength Grade A, B	AITM1-0002	MPa	dry	RT	103	100
					120	76	73
				wet	70	80	77
6b	In plane shear strength Grade C	AITM1-0002	MPa	dry	-55	75	73
					RT	91	89
					70	87	85
					120	69	67
				wet	RT	91	88
					70	75	70
					90	65	60
					120	50	48
6c	In plane shear modulus	AITM1-0002	GPa	dry	-55	5,9 ± 0,4	
					RT	5,4 ± 0,4	
					70	4,7 ± 0,4	
					120	3,9 ± 0,4	
				wet	RT	5,0 ± 0,4	
					70	4,0 ± 0,4	
					90	3,1 ± 0,4	
					120	1,5 ± 0,3	
(Continued)							

Table 5: Mechanical Properties (concluded)

No	Properties	Test method	Unit	Conditioning ^{b)}	Test temp. [°C]	Requirements ^{a)}	
						X	X _{min}
Material Code						AIMS05-27-002 A, B and C	
7	Interlaminar Shear Strength 0°	EN2563	MPa	dry	-55	115	110
					RT	88	83
					70	77	72
					120	60	57
				wet	RT	85	83
					70	60	57
					90	53	50
					120	40	38
8	G _{1C}	AITM1-0005	J/m ²	dry	RT	240	220
				wet	RT	210	190
9	G _{2C}	AITM1-0006	J/m ²	dry	RT	650	620
				wet	RT	550	430
10a	Interlaminar Shear Strength (reference)	EN2563 ^{c)}	MPa	dry	RT	90	85
				wet	70	68	65
10b	Interlaminar Shear Strength (skydrol, 70 °C/1000 h)	EN2563 ^{c)}	MPa	dry	70	-	-
				wet	70	> 68 ^{e)}	> 65 ^{e)}
10c	Interlaminar Shear Strength (skydrol/water, 50 °C/1000 h)	EN2563 ^{c)}	MPa	dry	70	-	-
				wet	70	> 68 ^{e)}	> 65 ^{e)}
10d	Interlaminar Shear Strength (fuel, RT/1000 h)	EN2563 ^{c)}	MPa	dry	70	-	-
				wet	70	> 68 ^{e)}	> 65 ^{e)}
10e	Interlaminar Shear Strength (MEK, RT/1h)	EN2563 ^{c)}	MPa	dry	RT	> 90 ^{f)}	> 85 ^{f)}
				wet	70	-	-
11	Resistance to strippers (Non Phenolic)	AITM1-0001 ^{d)}	%	dry	RT	> 0,9	> 0,9
				wet	70	-	-
^{a)} X means minimum average / X _{min} means minimum individual value. Requirements for Tensile, Compression, Plain, Open Hole and Filled Hole Tensile and Compression properties have been calculated using a theoretical thickness per ply of 0,125 mm for material code A 0,181 mm for Code B and 0,250 mm for material code C. Requirements for gross section. Requirements for other properties have been calculated with the real thickness.							
^{b)} Wet means 70 °C / 85 % R.H. according to EN2823 until saturation							
^{c)} Specimen lay-up according to AIMS05-27-000 Part 1.							
^{d)} According to AIMS05-27-000 Part 1 (to be precised in the QTP)							
^{e)} The values after conditioning will not have a decrease greater than the corresponding one in comparison with the wet (70 °C / 85 % R.H.) reference value at the equivalent temperature							
^{f)} Reference value is RT/Dry							

Table 6: Multiangular Mechanical Properties

No.	Properties	Test method	Unit	Cond. ^{b)}	Test temp. [°C]	Requirement ^{a)}	
						X	X _{min}
Material Code						AIMS05-27-002 C	
Lay-up: 10/80/10							
12a	Plain tensile strength	AITM1-0007 A (10/80/10)	MPa	dry	RT	470	440
				wet	70	440	410
12b	Plain tensile modulus	AITM1-0007 A (10/80/10)	MPa	dry	RT	34 ± 4	
				wet	70	34 ± 4	
12c	Open hole tensile strength	AITM1-0007 B (10/80/10)	MPa	dry	-55	320	310
					RT	320	300
				wet	70	280	260
12d	Plain compression strength	AITM1-0008 A (10/80/10)	MPa	dry	RT	440	400
				wet	70	340	300
12e	Plain compression modulus	AITM1-0008 A (10/80/10)	MPa	dry	RT	36 ± 3	
				wet	70	32 ± 3	
12f	Open hole compression strength	AITM1-0008 B (10/80/10)	MPa	dry	-55	270	260
					RT	240	220
				wet	70	180	160
12g	Filled hole compression strength	AITM1-0008 D (10/80/10)	MPa	wet	70	220	200
(Continued)							

(Continued)

Table 6: Multiangular Mechanical Properties

No.	Property	Test method	Unit	Cond. ^{b)}	Test temp. [°C]	Requirement ^{a)}	
						X	X _{min}
Material Code						AIMS05-27-002 A, B, C	
Lay-up: 25/50/25							
13a	Plain tensile strength	AITM1-0007 A (25/50/25)	MPa	dry	RT	760	725
				wet	70	760	725
13b	Plain tensile modulus	AITM1-0007 A (25/50/25)	MPa	dry	RT	60 ± 5	
				wet	70	60 ± 5	
13c	Open hole tensile strength	AITM1-0007 B (25/50/25)	MPa	dry	RT	490	460
				wet	70	490	460
					90	490	460
13d	Filled hole tensile strength	AITM1-0007 D (25/50/25)	MPa	dry	RT	445	425
				wet	70	425	405
					90	425	405
13e	Plain compression strength	AITM1-0008A (25/50/25)	MPa	dry	RT	600	510
				wet	70	430	360
13f	Plain compression modulus	AITM1-0008 A (25/50/25)	MPa	dry	RT	56 ± 6	
				wet	70	54 ± 6	
13g	Open hole compression strength	AITM1-0008 B (25/50/25)	MPa	dry	RT	300	280
				wet	70	235	200
					90	215	180
13h	Filled hole compression strength	AITM1-0008 D (25/50/25)	MPa	dry	RT	360	320
				wet	70	240	200
					90	200	160
13i	Compression after impact at 30J	AITM1-0010 (25/50/25)	MPa	dry	RT	250	230
				wet	70	200	180
(continued)							

(continued)

Table 6: Multiangular Mechanical Properties

No.	Property	Test method	Unit	Cond. ^{b)}	Test temp. [°C]	Requirement ^{a)}	
						X	X _{min}
Material Code						AIMS05-27-002 C	
Lay-up: 40/40/20							
14a	Plain Tensile Strength	AITM1-0007 A (40/40/20)	MPa	dry	-55	1050	1010
					RT	1160	1120
					70	1160	1120
				wet	70	1160	1120
14b	Plain Tensile Modulus	AITM1-0007 A (40/40/20)	MPa	dry	-55	84 ± 6	
					RT	84 ± 6	
					70	84 ± 6	
				wet	70	84 ± 6	
14c	Open hole tensile strength	AITM1-0007 B (40/40/20)	MPa	dry	-55	710	695
					RT	700	675
				wet	70	710	695
14d	Filled hole tensile strength	AITM1-0007 D (40/40/20)	MPa	dry	RT	610	580
					70	580	570
				wet	70	550	540
14e	Plain compression strength	AITM1-0008 A (40/40/20)	MPa	dry	-55	800	770
					RT	780	720
					70	730	700
				wet	70	560	500
14f	Plain compression Modulus	AITM1-0008 A (40/40/20)	MPa	dry	-55	72 ± 7	
					RT	74 ± 7	
					70	71 ± 7	
				wet	70	71 ± 7	
14g	Open hole compression strength	AITM1-0008 B (40/40/20)	MPa	dry	-55	430	405
					RT	400	375
				wet	70	300	275
(continued)							

(continued)

Table 6: Multiangular Mechanical Properties

No.	Properties	Test method	Unit	Cond. ^{b)}	Test temp. [°C]	Requirements ^{a)}	
						X	X _{min}
Material Code						AIMS05-27-002 C	
Lay-up: 40/40/20							
14h	Filled hole compression strength	AITM1-0008 D (40/40/20)	MPa	dry	RT	450	420
					70	330	310
				wet	70	300	280
Material Code						AIMS05-27-002 A, B	
Lay-up: 50/40/10							
15a	Plain Tensile Strength	AITM1-0007 A (50/40/10)	MPa	dry	-55	1450	1370
					RT	1500	1400
				wet	70	1600	1500
15b	Plain Tensile Modulus	AITM1-0007 A (50/40/10)	MPa	dry	-55	97 ± 7	
					RT	97 ± 7	
				wet	70	97 ± 7	
15c	Open hole tensile strength	AITM1-0007 B (50/40/10)	MPa	dry	-55	860	820
					RT	860	820
				wet	70	860	820
15d	Filled hole tensile strength	AITM1-0007 D (50/40/10)	MPa	dry	RT	680	640
					70	700	660
				wet	70	700	660
15e	Plain compression strength	AITM1-0008 A (50/40/10)	MPa	dry	-55	870	810
					RT	750	690
				wet	70	610	550
15f	Plain compression Modulus	AITM1-0008 A (50/40/10)	MPa	dry	-55	89 ± 8	
					RT	84 ± 8	
				wet	70	85 ± 8	
15g	Open hole compression strength	AITM1-0008 B (50/40/10)	MPa	dry	RT	430	390
				wet	70	355	315
15h	Filled hole compression strength	AITM1-0008 D (50/40/10)	MPa	dry	RT	440	400
					70	360	320
				wet	70	300	270
(continued)							

(continued)

Table 6: Multiangular Mechanical Properties

No.	Property	Test method	Unit	Cond. ^{b)}	Test temp. [°C]	Requirement ^{a)}	
						X	X _{min}
Material Code						AIMS05-27-002 C	
16a	Bearing Strength 2 % Elongation	AITM1-0009 10/80/10	MPa	dry	-55	830	770
					RT	850	790
				wet	70	600	500
16b	Bearing Strength Ultimate Load		MPa	dry	-55	960	920
					RT	1000	960
				wet	70	730	670
Material Code						AIMS05-27-002 A, B	
16c	Bearing Strength 2 % Elongation	AITM1-0009 25/50/25	MPa	dry	RT	930	850
				wet	70	730	690
					90	700	670
					120	530	510
16d	Bearing Strength Ultimate Load		MPa	dry	RT	950	910
				wet	70	750	700
					90	720	690
					120	580	540
Material Code						AIMS05-27-002 C	
16e	Bearing Strength 2 % Elongation	AITM1-0009 40/40/20	MPa	dry	-55	880	830
					RT	780	700
				wet	70	610	550
16f	Bearing Strength Ultimate Load		MPa	dry	-55	1040	990
					RT	980	950
				wet	70	720	690
(continued)							

(continued)

Table 6: Multiangular Mechanical Properties (concluded)

No.	Properties	Test method	Unit	Cond. ^{b)}	Test temp. [°C]	Requirements ^{a)}	
						X	X _{min}
Material Code						AIMS05-27-002 A, B	
16g	Bearing Strength 2 % Elongation	AITM1-0009 50/40/10	MPa	dry	-55	780	710
					RT	850	780
				wet	70	670	600
16h	Bearing Strength Ultimate Load		MPa	dry	-55	970	900
					RT	980	930
				wet	70	760	710
^{a)} X means minimum average / X _{min} means minimum individual value. Requirements for Tensile, Compression, Plain, Open Hole and Filled Hole Tensile and Compression properties have been calculated using a theoretical thickness per ply of 0,125 mm for material code A, 0,181 mm for Code B and 0,250 mm for material code C. Requirements for gross section. Requirements for other properties have calculated with the real thickness.							
^{b)} Wet means 70 °C / 85 % R.H. according to EN2823 until saturation.							

5 Designation

Refer to ABS5377.

6 Technical specification

The general technical requirements and responsibilities corresponding to this Material Specification are described in the Technical Specification AIMS05-27-000 Part 1 and Part 2.

7 Incoming Inspection

Materials qualified against this AIMS shall undergo "Technical Incoming Inspection" as per AIMS05-27-000 Part 100 prior to use.

Record of Revisions

Issue	Clause modified	Description of Modification
1 07.2009	-	New Standard
2 10.2019	3 4.1.3	AITM6-0011 added New sentence added on supplier's processing
3 03.2012	3 4.1.3	AITM6-0011 removed from reference table New wording
4 08.2012	Chapter 7 Tables 5 & 6	Chapter 7 "Incoming Inspection" added Revised requirements: Unidirectional Tensile Strength 70 °C/dry, 70 °C/wet and 90 °C/wet; Plain Tensile Strength 40/40/20 70 °C/dry and 70 °C/wet; Inter laminar Shear Strength -55 °C/dry; FHC and FHT 40/40/20 RT/dry
5 03.2016	Tables 5 & 6 Table 2	Revised requirements: Unidirectional Tensile Strength 70 °C/dry, 70 °C/wet and 90 °C/wet; PTM 40/40/20 and 25/50/25 all conditions; ILSS 0° 70 °C/wet and 90 °C/wet Work life requirements have been modified
6 06.2017	All Table 5	Tables reformatted as required Revised requirement: In plane shear strength (IPSS) Grade C 70 °C/wet and 90 °C/wet
7 10.2019	All Table 2 Table 3 Table 4 4.1.3 4.2.2 Table 5 Table 6	Updated to latest template and reformatted as required Note modified Note completed AITM3-0004 method C added Note 1 removed Table 7 reference removed CAI removed from Table 5, added in Table 6 CAI added to Table 6, Bearing test method reference updated, OHT requirement modified, Typo correction.
8 07.2021	Table 4 Table 5 & 6	CTE test method: test standard corrected. Note b modified Requirements updated to fit with test results and design values. Modified lines : 2a, 6b, 7, 12a, 12c, 13a, 13c,13d, 13g, 14a, 14c, 14d, 14g, 15a, 15g